PROJECT REPORT: AUTOMATED IT INCIDENT TICKET CLASSIFICATION USING SERVICENOW FLOW DESIGNER

1. INTRODUCTION

1.1 Project Overview

Educational institutions rely heavily on digital infrastructure for day-to-day operations. School IT helpdesks face high volumes of daily incident requests submitted by students, faculty, and administrative staff. These requests encompass recurring technical problems such as Wi-Fi connection drops, classroom projector failures, user account password lockouts, and slow workstation performance.

Currently, the institution's IT support pipeline relies on manual intervention by IT administrators to read incoming incident tickets, determine the problem domain, and assign an appropriate category and assignment group. This project introduces an automated ticket classification solution built natively within the **ServiceNow platform using Flow Designer** under the flow name **"Auto Classify School IT Tickets"**. The system inspects text fields (`short_description` and `description`) for pre-defined keyword patterns to auto-categorize, update records, and notify relevant teams upon submission.

1.2 Purpose

The primary objective of this project is to eliminate operational bottlenecks in the IT support service lifecycle. Specific goals include:

- **Automating Classification:** Eliminating the need for manual review of incoming incident tickets.
- **Reducing Mean Time to Resolution (MTTR):** Ensuring incidents reach the designated technical specialist instantly.
- **Standardizing Ticket Metadata:** Correctly applying categories (Network, Hardware, Password/Access, Performance/Software) to streamline helpdesk reporting.
- **Optimizing IT Staff Utilization:** Freeing support engineers from repetitive administrative triage so they can focus on resolving technical issues.

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives dozens of incident requests daily from students and teachers regarding Wi-Fi issues, projector failures, password resets, and slow computers. Currently, IT staff manually reviews each ticket to assign a category and support group, creating an administrative delay that increases resolution time and leads to unequal workload distribution among IT staff.

2.2 Empathy Map Canvas

Category	Description / Insights
Says	• Teachers: "*My classroom projector isn't working, and class starts in 5 minutes!*" • Students: "*I submitted a password reset ticket hours ago and haven't heard back.*" • IT Staff: "*We spend the first hour of every day just categorizing and assigning incoming tickets.*"
Thinks	• Users: "*The IT helpdesk takes too long to respond to simple problems.*" • IT Staff: "*There must be an automated way to route simple issues like password resets automatically.*"
Does	• Users log vague descriptions (e.g., "*Internet not working*" or "*PC slow*"). • IT administrators manually read each record line by line and adjust dropdown fields in ServiceNow.
Feels	• Teachers & Students feel frustrated by unexpected technical delays during academic hours. • IT staff feel overwhelmed by routine admin work and overburdened during peak request times.
Pains	• High initial response lag. • Inconsistent ticket categorization due to human error. • Delays in urgent classroom tech fixes (e.g., projectors & campus Wi-Fi).
Gains	• Instant automated routing and category tagging. • Reduced triage time from hours to sub-100 milliseconds. • Faster resolution times for critical classroom hardware and account access issues.

2.3 Brainstorming

During the ideation phase, three key architectural approaches were evaluated:

1. **Custom JavaScript Business Rules:** Traditional script-based logic attached to the Incident table. While functional, it requires maintenance by software developers and is difficult to audit visually.
2. **External Machine Learning API:** Sending incident text to an external NLP model via REST integrations. While intelligent, this adds cloud latency, licensing costs, and data privacy concerns for student data.
3. **ServiceNow Flow Designer with Keyword Rules (Selected Solution):** A native, low-code workflow engine embedded inside ServiceNow. It allows sub-second keyword mapping across `short_description` and `description` fields without external dependencies or heavy code maintenance.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System / Touchpoint	User Pain Point	Automated Workflow Experience
1. Issue Occurrence	Wi-Fi drops / Projector fails / Password forgotten / PC slow.	Campus Classroom / Workstation	Interrupted lecture session or portal lockout.	N/A
2. Ticket Submission	User fills portal form with incident details.	ServiceNow Service Portal	Delays if ticket sits in unassigned queue.	Ticket created with Trigger: `Incident WorkFlow Created`.
3. Triage & Classification	System evaluates sequential logic conditions.	ServiceNow Flow Designer Engine	Old System: Wait for manual review by IT staff.	**Automated:** Target condition evaluates **True** in 1ms to 6ms.
4. Assignment & Routing	System updates record metadata & emails team.	ServiceNow Incident Management	Misrouted tickets causing delays.	Update Record & Send Email execute in sub-100ms time.
5. Resolution	Technician receives instant alert and resolves issue.	On-Site / Network / IT Support	Prolonged downtime.	Total execution completed between 5ms and 81ms.

3.2 Solution Requirement

Functional Requirements:

- **FR-01 (Trigger):** System must automatically launch `Auto Classify School IT Tickets` upon record insertion on the Incident table.
- **FR-02 (Text Parsing & Flow Logic):**
  - **Condition 1:** IF Short Description contains *Wifi* or *Network* → Update Record & Send Email.
  - **Condition 2:** ELSE IF Short Description contains *Projector* or *Hardware* → Update Record.
  - **Condition 3:** ELSE IF Short Description contains *Forgot password* → Update Record & Send Email.
  - **Condition 4:** ELSE IF Short Description contains *Performance* → Update Record & Send Email.
- **FR-03 (Notification):** Send automated email notification upon triggering true conditional matches.

Non-Functional Requirements:

- **NFR-01 (Performance Execution):** Flow execution must complete within sub-second thresholds (< 100 ms).
- **NFR-02 (Reliability):** Zero error state recorded under System Administrator execution context across all test cases.

3.3 Technology Stack

- **Platform:** ServiceNow Enterprise Instance
- **Workflow Automation Engine:** ServiceNow Flow Designer
- **Flow Name:** `Auto Classify School IT Tickets`
- **Execution Context:** System Administrator

4. PROJECT DESIGN

4.1 Proposed Solution & Decision Architecture

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1. IF Short Description contains "Wifi" OR "Network"
       -> Core Action: Update Record (Completed: 5ms)
       -> Core Action: Send Email (Completed: 40ms)
2. ELSE IF Short Description contains "Projector" OR "Hardware"
       -> Core Action: Update Record (Completed: 2ms)
3. ELSE IF Short Description contains "Forgot password"
       -> Core Action: Update Record (Completed: 3ms)
       -> Core Action: Send Email (Completed: 32ms)
4. ELSE IF Short Description contains "Performance"
       -> Core Action: Update Record (Completed: 1ms)
       -> Core Action: Send Email (Completed: 78ms)
```

5. PROJECT PLANNING & SCHEDULING

Phase	Task Description	Duration	Key Deliverables
Phase 1: Requirements	Identify school helpdesk issue types and gather historical incident data.	Week 1	Keyword Dictionary & Requirement Specification
Phase 2: Environment Setup	Configure ServiceNow Instance and Incident Table choices.	Week 2	Configured Choice Lists & Data Models
Phase 3: Flow Construction	Build `Auto Classify School IT Tickets` flow, trigger, and logic steps.	Week 3	Active Flow Designer Definition
Phase 4: Testing & QA	Execute test runs in ServiceNow Flow Designer canvas across all domains.	Week 4	Test Execution Logs & Performance Diagnostics
Phase 5: Deployment	Publish flow to production and monitor ticket routing accuracy.	Week 5	Fully Operational Automated Triage System

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 ServiceNow Flow Test Execution Master Matrix

Step / Action	Test 1: Wifi / Network (09:15:40)	Test 2: Projector (06:36:10)	Test 3: Forgot Password (06:38:44)	Test 4: Performance (07:08:18)
Trigger	Incident WorkFlow Created	Incident WorkFlow Created	Incident WorkFlow Created	Incident WorkFlow Created
1. Wifi / Network	**Evaluated - True (6 ms)**	Evaluated - False (1 ms)	Evaluated - False (2 ms)	Evaluated - False (2 ms)
2. Update Record	**Completed (5 ms)**	Not Run	Not Run	Not Run
3. Projector / Hardware	Not Run	**Evaluated - True (2 ms)**	Evaluated - False (1 ms)	Evaluated - False (0 ms)
4. Update Record	Not Run	**Completed (2 ms)**	Not Run	Not Run
5. Forgot Password	Not Run	Not Run	**Evaluated - True (4 ms)**	Evaluated - False (0 ms)
6. Update Record	Not Run	Not Run	**Completed (3 ms)**	Not Run
7. Performance	Not Run	Not Run	Not Run	**Evaluated - True (1 ms)**
8. Update Record	Not Run	Not Run	Not Run	**Completed (1 ms)**
9. Send Email	**Completed (40 ms)**	Not Run	**Completed (32 ms)**	**Completed (78 ms)**
Total Execution Time	**51 ms** (**PASS**)	**~5 ms** (**PASS**)	**42 ms** (**PASS**)	**81 ms** (**PASS**)

7. RESULTS

7.1 Complete Execution Diagnostics Log

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=== TEST RUN 1: WIFI / NETWORK === Flow Name: Auto Classify School IT Tickets Run State: Completed | Run As: System Administrator Timestamp: 2026-10-01 09:15:40 | Total Time: 51ms | Errors: 0 [09:15:40] Trigger: Incident WorkFlow Created -> Fired [09:15:40] Step 1: If SD is Wifi or Network -> Evaluated TRUE (6ms) [09:15:40] Step 2: Update Record -> COMPLETED (5ms) [09:15:40] Step 9: Send Email -> COMPLETED (40ms) [09:15:40] Steps 3-8: Skipped / Not Run (Rule matched at Step 1) === TEST RUN 2: PROJECTOR / HARDWARE === Flow Name: Auto Classify School IT Tickets Run State: Completed | Run As: System Administrator Timestamp: 2026-10-01 06:36:10 | Total Time: ~5ms | Errors: 0 [06:36:10] Trigger: Incident WorkFlow Created -> Fired [06:36:10] Step 1: If SD is Wifi or Network -> Evaluated FALSE (1ms) [06:36:10] Step 3: Else If SD is Projector or Hardware -> Evaluated TRUE (2ms) [06:36:10] Step 4: Update Record -> COMPLETED (2ms) [06:36:10] Steps 5-9: Skipped / Not Run (Rule matched at Step 3) === TEST RUN 3: FORGOT PASSWORD === Flow Name: Auto Classify School IT Tickets Run State: Completed | Run As: System Administrator Timestamp: 2026-10-01 06:38:44 | Total Time: 42ms | Errors: 0 [06:38:44] Trigger: Incident WorkFlow Created -> Fired [06:38:44] Step 1: If SD is Wifi or Network -> Evaluated FALSE (2ms) [06:38:44] Step 3: Else If SD is Projector or Hardware -> Evaluated FALSE (1ms) [06:38:44] Step 5: Else If SD is Forgot password -> Evaluated TRUE (4ms) [06:38:44] Step 6: Update Record -> COMPLETED (3ms) [06:38:44] Step 9: Send Email -> COMPLETED (32ms) === TEST RUN 4: PERFORMANCE === Flow Name: Auto Classify School IT Tickets Run State: Completed | Run As: System Administrator Timestamp: 2026-10-01 07:08:18 | Total Time: 81ms | Errors: 0 [07:08:18] Trigger: Incident WorkFlow Created -> Fired [07:08:18] Step 1: If SD is Wifi or Network -> Evaluated FALSE (2ms) [07:08:18] Step 3: Else If SD is Projector or Hardware -> Evaluated FALSE (0ms) [07:08:18] Step 5: Else If SD is Forgot password -> Evaluated FALSE (0ms) [07:08:18] Step 7: Else If SD is Performance -> Evaluated TRUE (1ms) [07:08:18] Step 8: Update Record -> COMPLETED (1ms) [07:08:18] Step 9: Send Email -> COMPLETED (78ms)
```

8. ADVANTAGES & DISADVANTAGES

Advantages:

- **Sub-100ms High-Speed Execution:** Processes complete classification, record updates, and emails consistently between **5 ms and 81 ms**.
- **Early Decision Termination:** Matches top-tier conditions (like Wi-Fi or Projectors) early in 5 ms to 51 ms without processing lower branches.
- **Zero Maintenance Scripting:** Built entirely using low-code visual blocks, minimizing technical debt and scripting errors.

Disadvantages:

- **Sequential Evaluation Order:** Stops evaluating once the first true condition matches.
- **Keyword Matching Dependency:** Relies on standardized short description keywords submitted by users.

9. CONCLUSION

The **Auto Classify School IT Tickets** flow successfully automates IT helpdesk ticket processing for educational institutions. As proven across all four production test runs (Wi-Fi, Projector, Password Reset, and Performance), incoming incident requests are evaluated, updated, and dispatched with an average total latency of under 45 milliseconds. This eliminates manual triage delays and ensures rapid response times for critical campus IT services.

10. FUTURE SCOPE

- **ServiceNow Predictive Intelligence:** Integrate Machine Learning models for intent recognition on complex user queries.
- **Virtual Agent Integration:** Automatically present password reset KB articles before ticket submission.
- **Dynamic On-Call Workload Balancing:** Direct tickets automatically to logged-in technicians based on live work schedules.